

12_May-Nov_2020 • Unit 1

1H + 2H source-order past paper

Dr Eslam Ahmed

eliteigcse.com | WhatsApp +20 112 000 9622

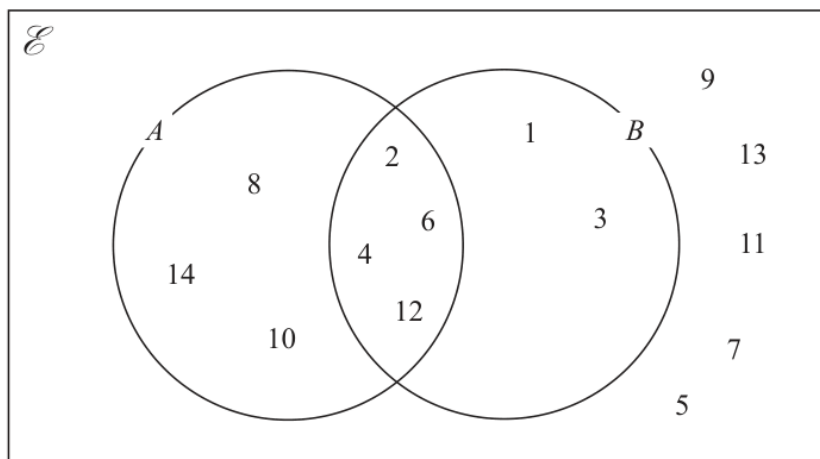
L-to-M | CONVERTED MODULAR EDITION

Selected from Linear Higher papers; not an original Modular examination paper.

Contents

- 1H Q01 • Probability Diagrams - Venn & Tree Diagrams • 4 marks
- 1H Q02 • Probability Toolkit • 4 marks
- 1H Q03 • Linear Graphs ($y = mx + c$) • 3 marks
- 1H Q06 • Standard & Compound Units • 3 marks
- 1H Q07 • Factorising • 8 marks
- 1H Q11 • Area & Perimeter • 6 marks
- 1H Q15 • Probability Diagrams - Venn & Tree Diagrams • 5 marks
- 1H Q17 • Rounding, Estimation & Bounds • 4 marks
- 1H Q18 • Histograms • 5 marks
- 1H Q19 • Surds • 3 marks
- 1H Q21 • Algebraic Fractions • 5 marks
- 1H Q22 • Coordinate Geometry • 4 marks
- 2H Q03 • Angles in Polygons & Parallel Lines • 5 marks
- 2H Q06 • Linear Graphs ($y = mx + c$) • 2 marks
- 2H Q07 • Right-Angled Triangles - Pythagoras & Trigonometry • 3 marks
- 2H Q08 • Area & Perimeter • 4 marks
- 2H Q12 • Algebraic Roots & Indices • 3 marks
- 2H Q13 • Fractions, Decimals & Percentages • 2 marks
- 2H Q14 • Set Notation & Venn Diagrams • 6 marks
- 2H Q18 • 3D Pythagoras & Trigonometry • 5 marks
- 2H Q20 • Combined & Conditional Probability • 5 marks

- 1 The numbers from 1 to 14 are shown in the Venn diagram.



- (a) List the members of the set $A \cap B$

.....
(1)

- (b) List the members of the set B'

.....
(1)

A number is picked at random from the numbers in the Venn diagram.

- (c) Find the probability that this number is in set A but is **not** in set B .

.....
(2)

(Total for Question 1 is 4 marks)

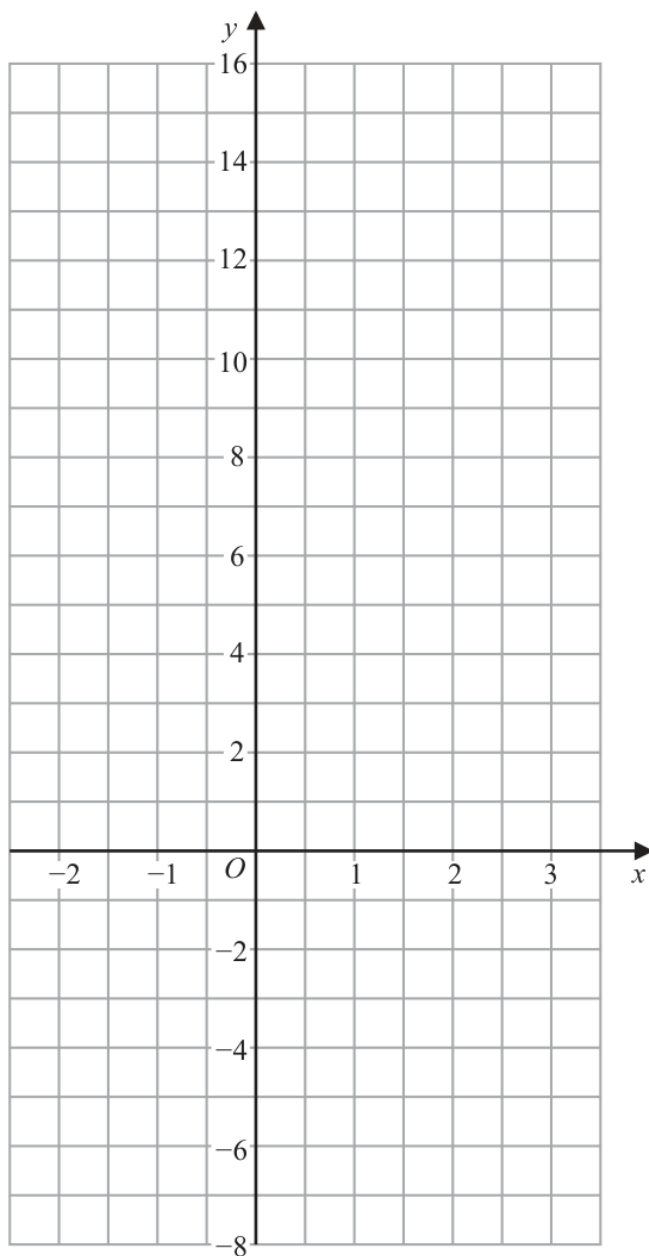
- 2 Toy cars are made in a factory.
The toy cars are made for 15 hours each day.
5 toy cars are made every 12 seconds.

For the toy cars made each day, the probability of a toy car being faulty is 0.002

Work out an estimate of the number of faulty toy cars that are made each day.

.....
(Total for Question 2 is 4 marks)

- 3 On the grid, draw the graph of $y = 7 - 4x$ for values of x from -2 to 3



(Total for Question 3 is 3 marks)

- 6 A rocket travelled 100 km at an average speed of 28 440 km/h.

Work out how long it took the rocket to travel the 100 km.

Give your answer in seconds, correct to the nearest second.

..... seconds

(Total for Question 6 is 3 marks)

- 7 (a) Solve $5(4 - x) = 7 - 3x$
Show clear algebraic working.

$$x = \dots\dots\dots$$

(3)

- (b) Factorise fully $16m^3g^3 + 24m^2g^5$

$$\dots\dots\dots$$

(2)

- (c) (i) Factorise $y^2 - 2y - 48$

$$\dots\dots\dots$$

(2)

- (ii) Hence, solve $y^2 - 2y - 48 = 0$

$$\dots\dots\dots$$

(1)

(Total for Question 7 is 8 marks)

- 11 The diagram shows trapezium $ABCD$ in which BC and AD are parallel.

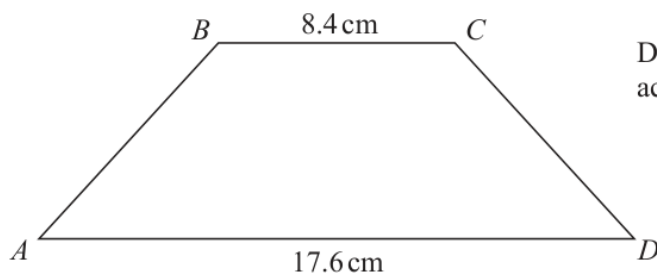


Diagram **NOT**
accurately drawn

The trapezium has exactly one line of symmetry.

$$BC = 8.4 \text{ cm}$$

$$AD = 17.6 \text{ cm}$$

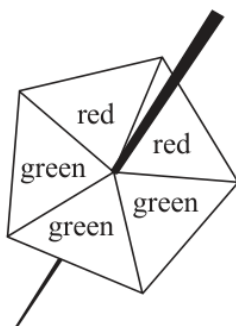
The trapezium has area 179.4 cm^2

Work out the size of angle ABC .

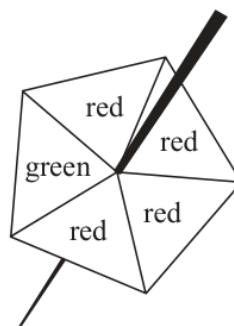
Give your answer correct to 1 decimal place.

(Total for Question 11 is 6 marks)

15 Harry has two fair 5-sided spinners.



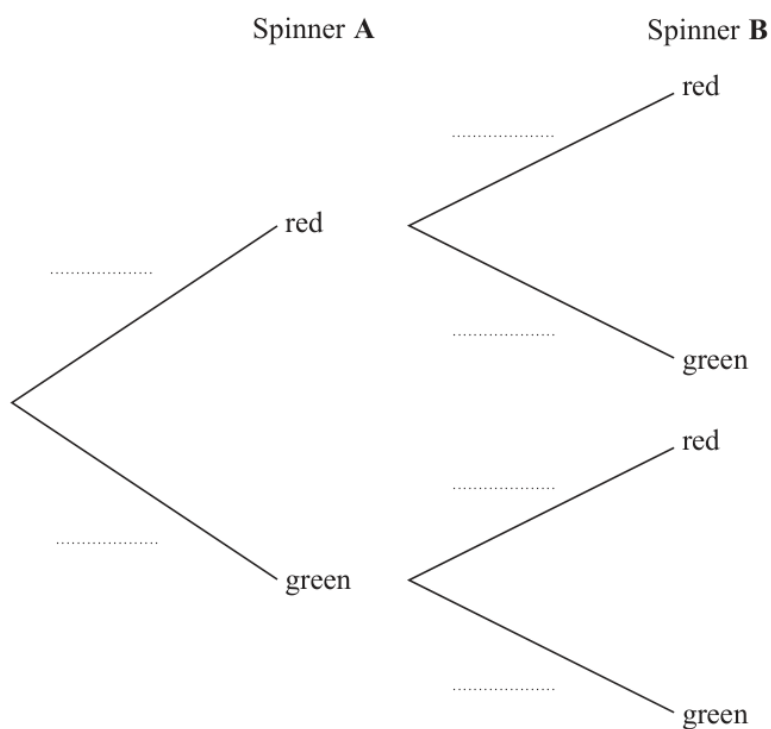
Spinner A



Spinner B

Harry is going to spin each spinner once.

(a) Complete the probability tree diagram.



(2)



(b) Work out the probability that at least one of the spinners will land on green.

.....
(3)

(Total for Question 15 is 5 marks)

- 17 A metal block has a mass of 5 kg, correct to the nearest 50 grams.
The block has a volume of $(1.84 \times 10^{-3}) \text{ m}^3$, correct to 3 significant figures.

Work out the upper bound for the density of the block.
Give your answer in kg/m^3 correct to 1 decimal place.
Show your working clearly.

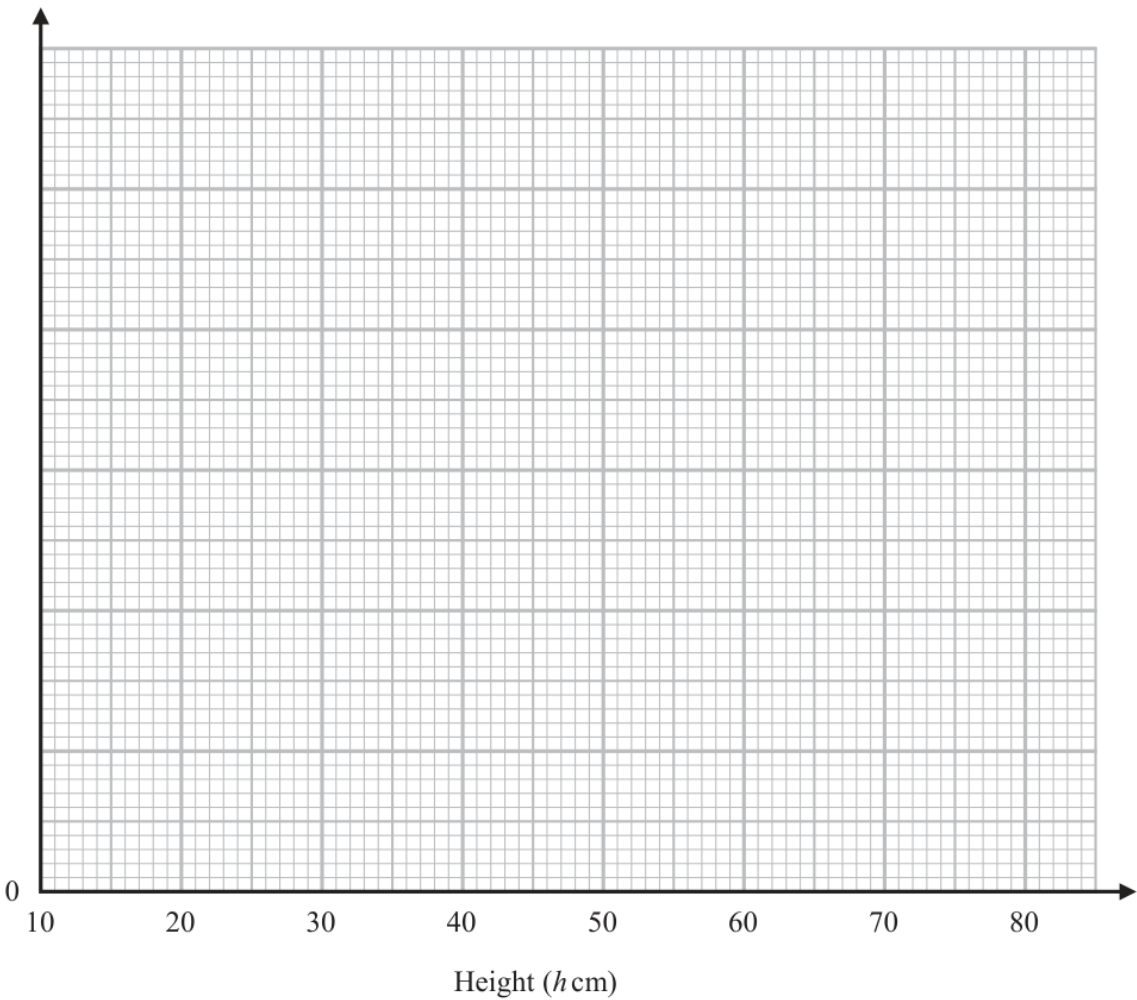
..... kg/m^3

(Total for Question 17 is 4 marks)

18 The table gives information about the heights, in centimetres, of some plants.

Height (h cm)	Frequency
$10 < h \leq 20$	35
$20 < h \leq 35$	45
$35 < h \leq 50$	75
$50 < h \leq 70$	40
$70 < h \leq 80$	8

(a) On the grid, draw a histogram for this information.



(3)



(b) Work out an estimate for the number of these plants with a height greater than 40 cm.

.....
(2)

(Total for Question 18 is 5 marks)

- 19 Without using a calculator, rationalise the denominator of $\frac{6}{3 - \sqrt{7}}$
- Simplify your answer.
You must show each stage of your working.

(Total for Question 19 is 3 marks)

21 Express

$$\frac{1}{3x-2} \times \frac{9x^2-4}{3x^2-13x-10} - \frac{7}{x-1}$$

as a single fraction in its simplest form.

(Total for Question 21 is 5 marks)

22 $ABCD$ is a rhombus.

The diagonals, AC and BD , intersect at the point M .

The coordinates of M are $(6, -11)$

The points A and C both lie on the line with equation $2y + 7x = 20$

Find the exact coordinates of the point where the line through B and D intersects the y -axis.

(..... ,)

(Total for Question 22 is 4 marks)

3

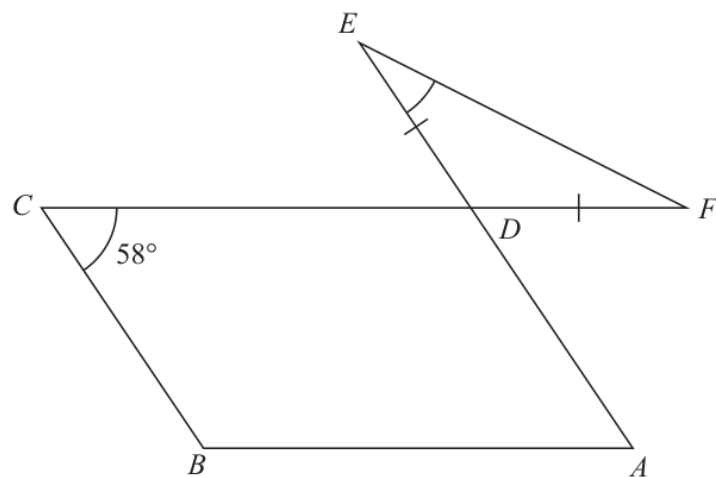


Diagram **NOT**
accurately drawn

The diagram shows a parallelogram $ABCD$ and an isosceles triangle DEF in which $DE = DF$

CDF and ADE are straight lines.

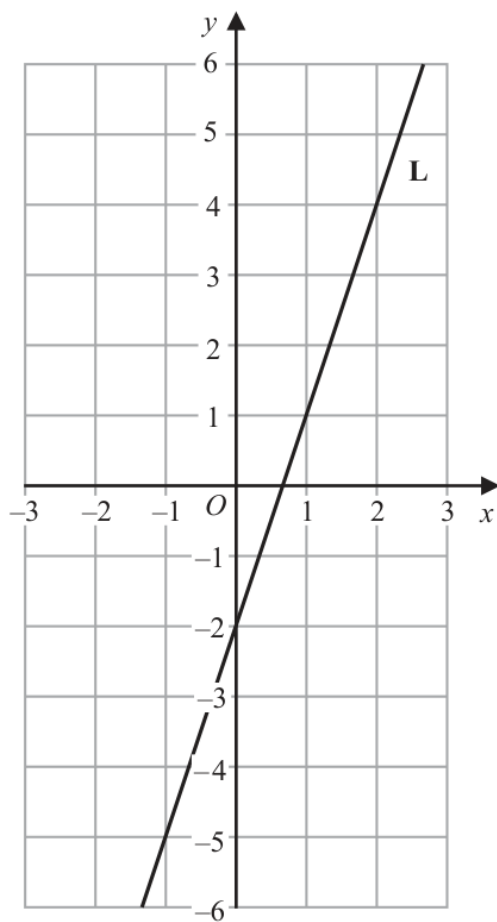
Angle $BCD = 58^\circ$

Work out the size of angle DEF .

Give a reason for each stage of your working.

(Total for Question 3 is 5 marks)

- 6 The line **L** is shown on the grid.



Find an equation for **L**.

(Total for Question 6 is 2 marks)

- 7 The diagram shows a right-angled triangle.

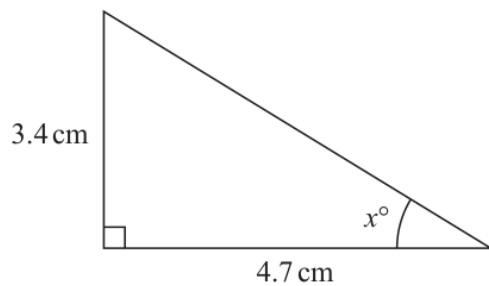


Diagram **NOT**
accurately drawn

Calculate the value of x .
Give your answer correct to one decimal place.

$x =$

(Total for Question 7 is 3 marks)

- 8 The diagram shows an isosceles triangle.

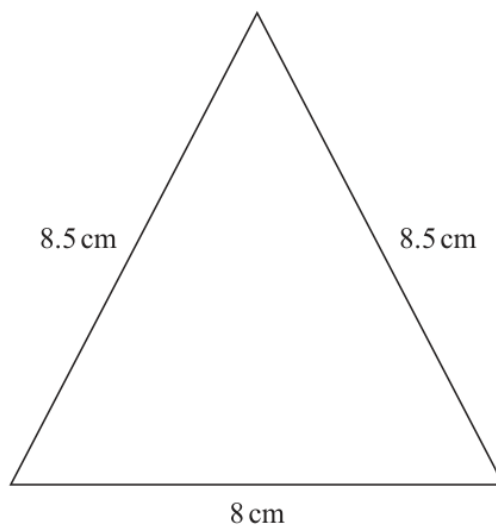


Diagram **NOT**
accurately drawn

Work out the area of the triangle.

.....cm²

(Total for Question 8 is 4 marks)

12 Given that $\frac{3^x}{9^{3x}} = 81$

find the value of x .

Show clear algebraic working.

$$x = \dots\dots\dots$$

(Total for Question 12 is 3 marks)

13 Use algebra to show that $0.\dot{6}8\dot{1} = \frac{15}{22}$

(Total for Question 13 is 2 marks)

14 $\mathcal{E} = \{\text{integers } x \text{ such that } 10 \leq x \leq 25\}$

$$A = \{x : x < 18\}$$

$$B = \{x : 13 \leq x < 22\}$$

(a) Write down $n(A)$

(1)

(b) List the members of the set $(A \cup B)'$

(2)

(c) List the members of the set $A' \cap B$

(2)

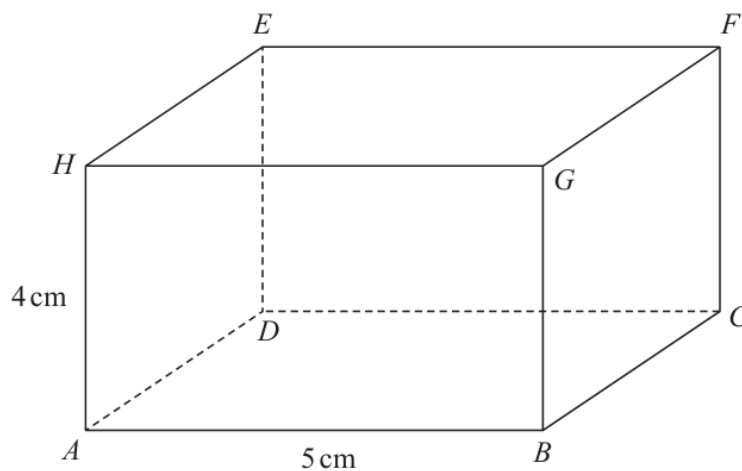
$$C \subset A, C \subset B \text{ and } n(C) = 5$$

(d) List the members of the set C

(1)

(Total for Question 14 is 6 marks)

18 The diagram shows cuboid $ABCDEFGH$.



$$AB = 5 \text{ cm}$$

$$AH = 4 \text{ cm}$$

The size of the angle between CH and the plane $ABCD$ is 35°

Calculate the volume of the cuboid.

Give your answer correct to 3 significant figures.

.....cm³

(Total for Question 18 is 5 marks)

20 A bag contains X counters.

There are only red counters and blue counters in the bag.

There are 4 more blue counters than red counters in the bag.

Finty takes at random 2 counters from the bag.

The probability that Finty takes 2 blue counters from the bag is $\frac{3}{8}$

Work out the value of X .

Show clear algebraic working.

(Total for Question 20 is 5 marks)